

## System thinking —

We have seen 4 basic operations on our signals. But now the question is how these transformations are applied on our signals. The answer is — it depends on what type of signal we are working with. For example, if you have attended CSE251, you have seen op-amp based signal manipulation for electrical signals. If you have attended CSE461 you have seen mechanical control systems for mechanical signals (hopefully). So the devices that manipulate signals vary a lot and specific domains can have their own standalone courses. But the best part is, we can mathematically model these devices for abstraction. In standard representation, we call these devices, **systems**.

**Takeaway** — This is an abstract course, so we will focus on domain agnostic signals & systems. We will only study the mathematical modelling (at best) and not deal with how to build these systems irl.

For this segment of the course we will simply represent systems as descriptive blocks.

For example —

$$x[n] \rightarrow \boxed{\text{Time shifting, } K} \rightarrow x[n-K]$$

$$x[n] \rightarrow \boxed{\text{Time scaling, } \beta} \rightarrow x[\beta n]$$

$$x[n] \rightarrow \boxed{\text{Amp. shifting, } c} \rightarrow x[n] + c$$

$$x[n] \rightarrow \boxed{\text{Amp. scaling, } \alpha} \rightarrow \alpha x[n]$$

Amp. shifting can also be represented as —

$$x[n] \rightarrow \bigcirc + \rightarrow x[n] + c$$

$\uparrow$   
 $c$

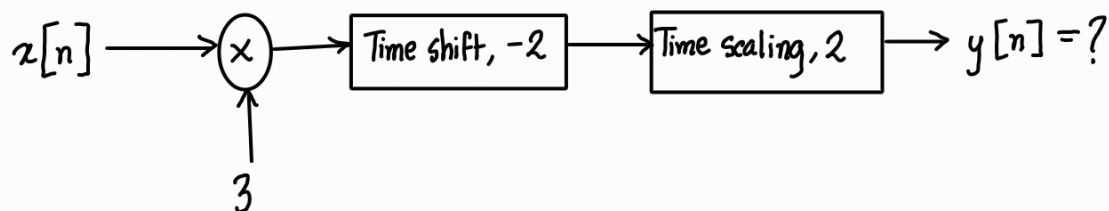
Amp. scaling can be represented as —

$$x[n] \rightarrow \bigcirc \times \rightarrow \alpha x[n]$$

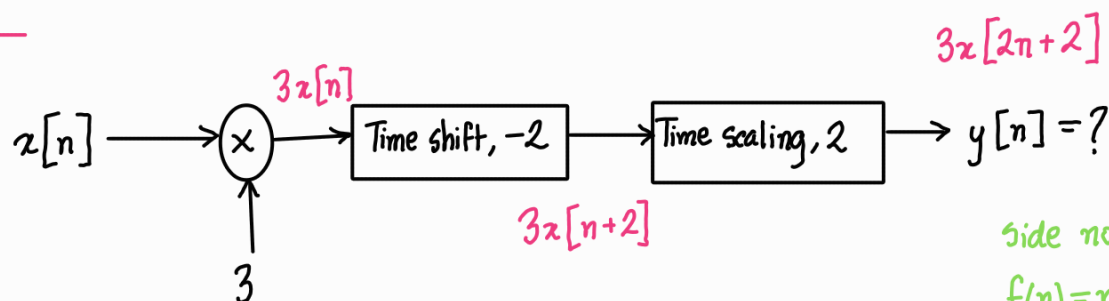
$\uparrow$   
 $\alpha$

## Multiple transformations —

Consider a system —

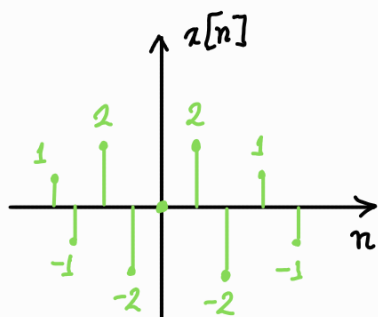


Solving —



side note —  
 $f(n) = n+2$   
 then  $f(2n) = 2n+2$

Now if I give you the waveshape of  $z[n]$ , can you draw  $y[n]$ ?



Draw  $y[n]$ .

Now clearly its very hard to draw it directly from  $y[n] = 3z[2n+2]$ . But,

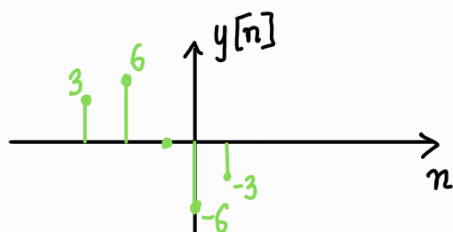
We can go step by step —

after amp. scaling —  $3z[n] = \{3, -3, 6, -6, 0, 6, -6, 3, -3\}$  ←  $inter_1[n] = 3z[n]$

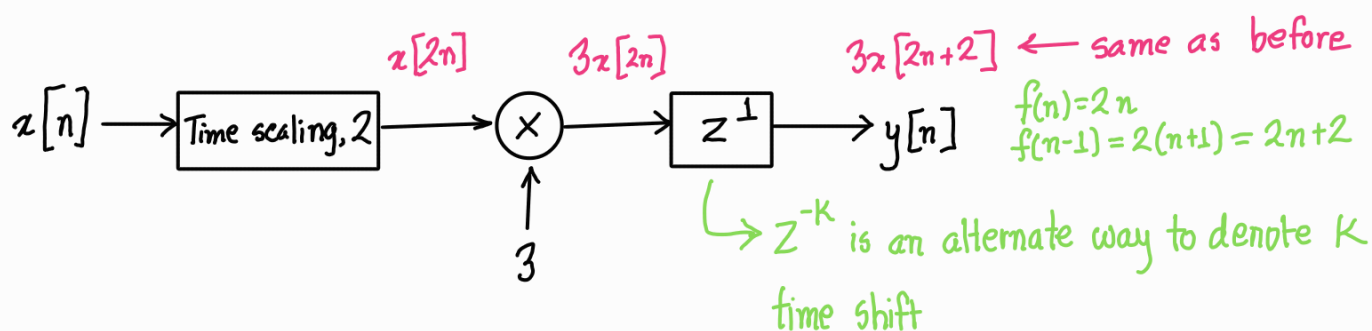
after time shifting —  $3z[n+2] = \{3, -3, 6, -6, 0, 6, -6, 3, -3\}$  ←  $inter_2[n] = inter_1[n+2]$

after time scaling —  $3z[2n+2] = \{3, 6, 0, -6, -3\}$  ←  $y[n] = inter_2[2n]$

So,  $y[n]$  in drawing —



Systems are generally not unique. For example let's try a new system —



Sanity check —

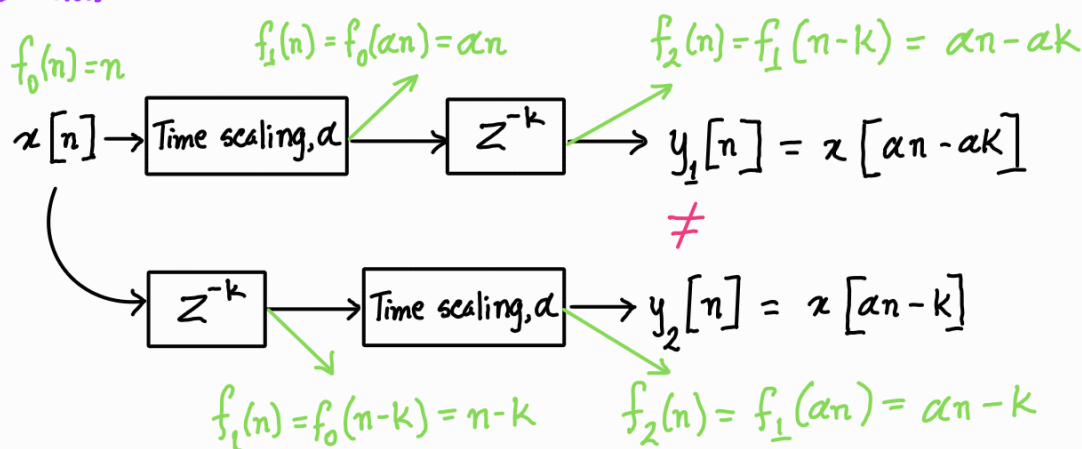
$$x[n] = \{1, -1, 2, -2, 0, 2, -2, 1, -1\}$$

$$\text{inter}_1[n] = x[2n] = \{1, 2, 0, -2, -1\}$$

$$\text{inter}_2[n] = 3 \cdot \text{inter}_1[n] = \{3, 6, 0, -6, -3\}$$

$$\text{inter}_3[n] = \text{inter}_2[n+1] = \{3, 6, 0, -6, -3\} \leftarrow y[n], \text{ same as before}$$

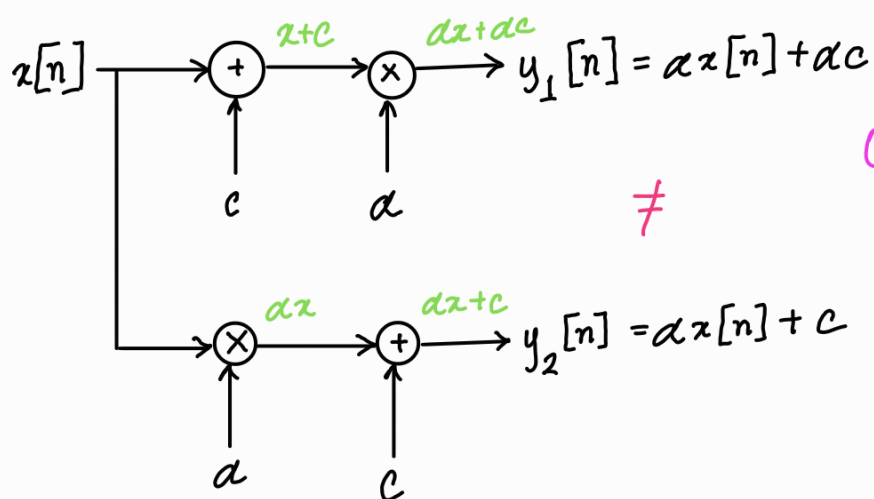
**Cautions —**



When in confusion always break into functions.

**NB —** This inequality occurs because all operations are done on  $n$ .

We can observe a similar behavior for amp. manipulation.



Operations are more intuitive though

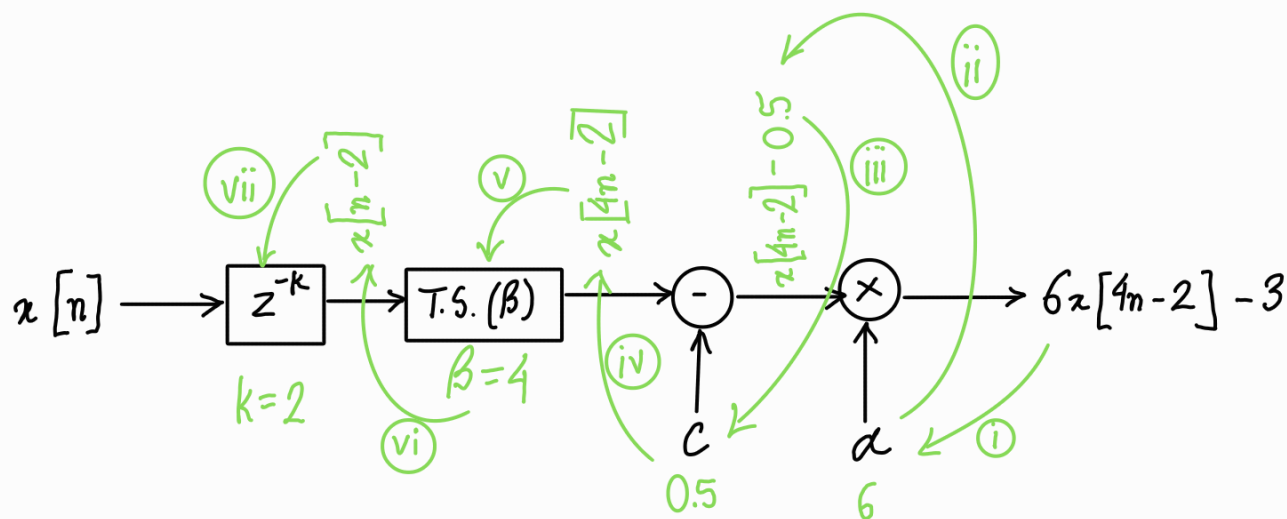
### Takeaway —

amp. manipulation works on only the  $x$  part and time manipulation only works on the  $n$  part. Time operations are special because they are applied wrt  $n$ .

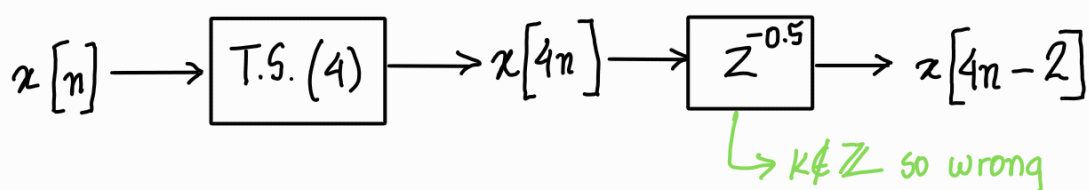
Now with this knowledge can you work backwards?

Try to create a system that takes an input  $x[n]$  and generates  $y[n] = 6x[4n-2] - 3$

Note that in  $6x[4n-2] - 3$  we have all 4 operations, so we need all the blocks.



In this case  $T.S.$  and  $z^{-k}$  blocks cannot be interchanged even by adjusting values because if we exchanged them, we would get something like —



Now, let's build more on this concept. Let's say  $x[n] = \{1, 2, -3, -4, 4, 3, -2, -1\}$

Then what will be  $6x[4n-2] - 3$ ?

Since we have already built the system, let's go one step at a time at first and then I will show you a shortcut where you will not have to build the system.

$$x[n] = \{1, 2, -3, -4, 4, 3, -2, -1\}$$

$$x[n-2] = \{1, 2, -3, -4, 4, 3, -2, -1\}$$

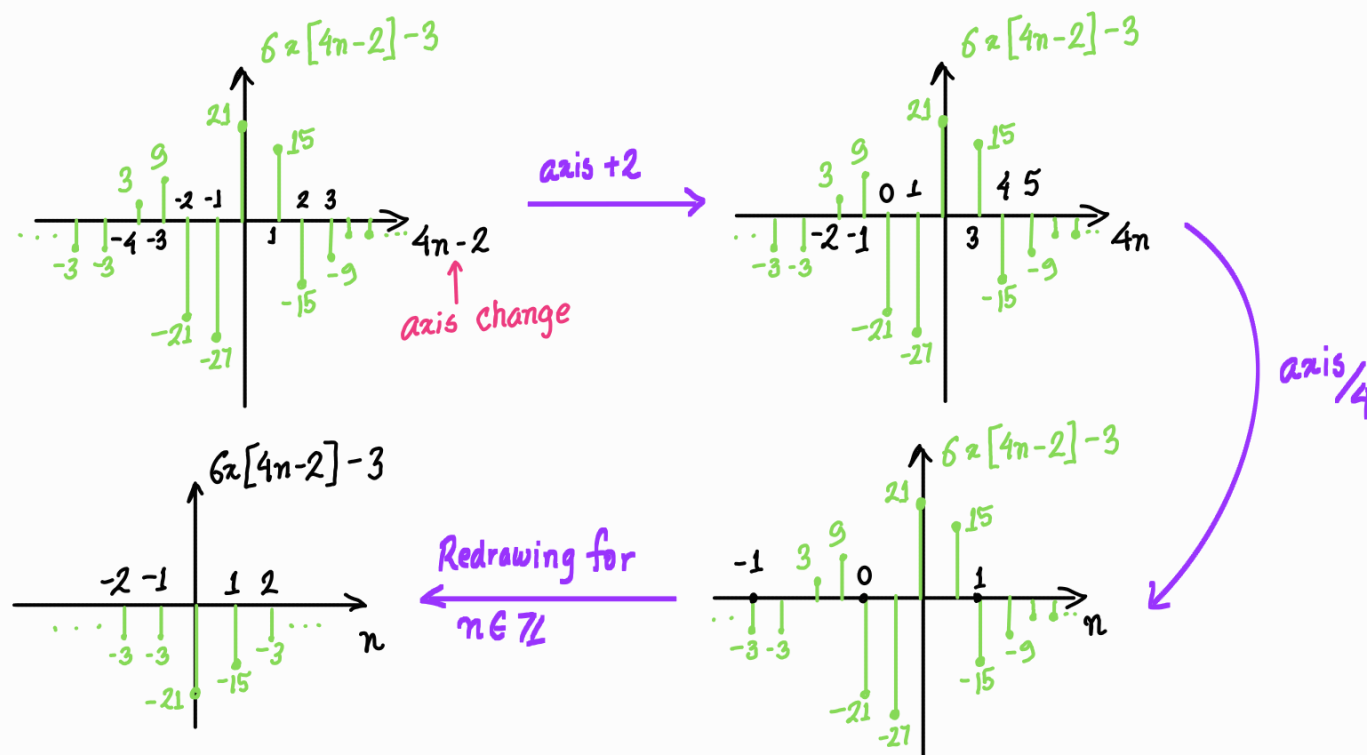
$$x[4n-2] = \{-3, -2\}$$

$$x[4n-2] - 0.5 = \{\dots, -0.5, -3.5, -2.5, -0.5, \dots\}$$

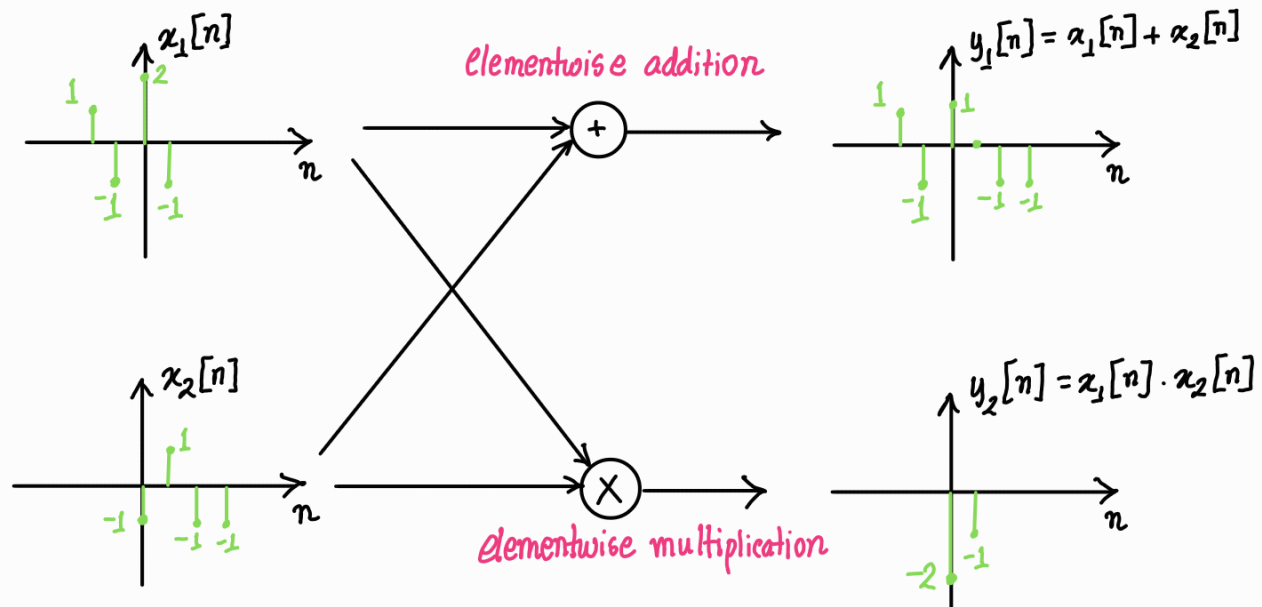
$$6x[4n-2] - 3 = \{\dots, -3, -21, -15, -3, \dots\}$$

Now the shortcut —

handle all the amp. manipulations at first —



So far, in terms of amp. manipulation, we have combined signals with constants. But can we combine signals with signals? We can. For example —



In the next lecture we will study more about systems and we will find a way to define a system as a signal.